

GE Healthcare

Product Network Filters (PNF)

IRIX version 3.0



OPERATING DOCUMENTATION

5197408
Revision 1

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Important Information

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A' CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。

GEMS 以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本。

非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

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Revision History

Rev	Date	Reason For Change
1	12/07/2006	Initial Release

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Table of Contents

1 Product Network Filters for IRIX 11

 1.1 Overview 11

 1.2 Firewall State 11

 1.3 Firewall Settings 12

 1.4 Tabs 14

 1.4.1 Named Services Tab 14

 1.4.2 Allowed Nodes Tab 16

 1.4.3 DICOM Tab 18

 1.4.4 Expert Tab 19

 1.5 Troubleshooting 21

 1.6 Finalization 22

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1 Product Network Filters for IRIX

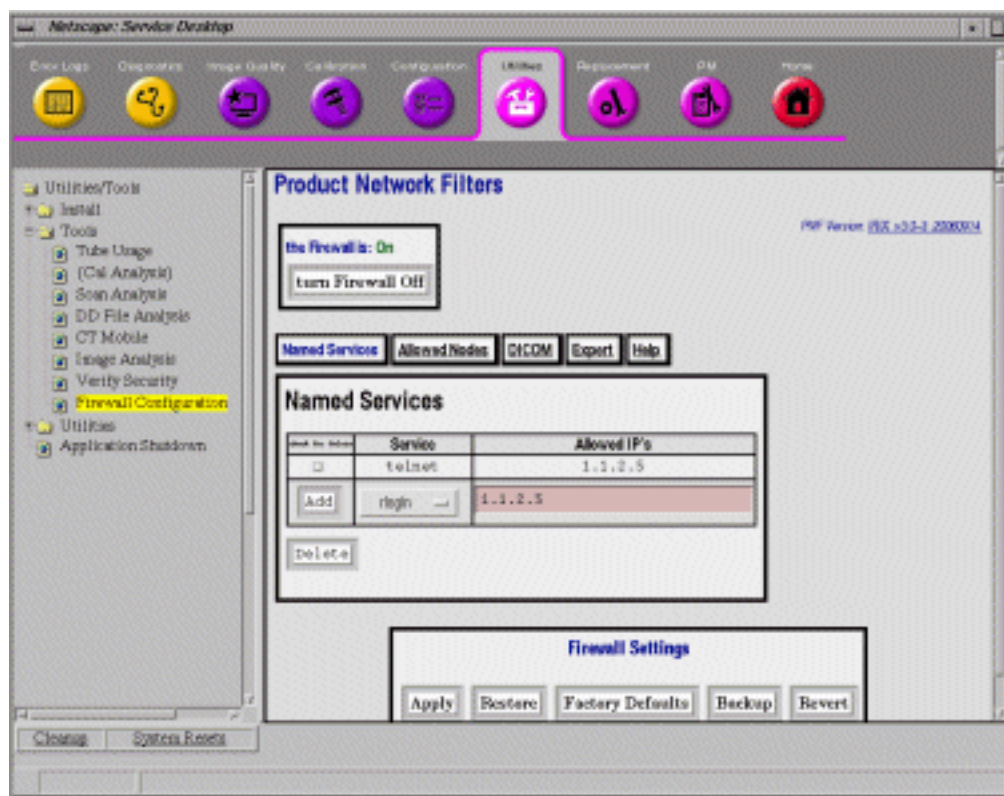
Last Revised:	12/07/2006
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1.1 Overview

Product Network Filters (PNF) is a software firewall tool which allows a site Network Administrator to control traffic from the site's internal network to the Diagnostic Imaging System. This tool was created in response to the customer's need for security.

This tool is accessible from the Common Service Desktop (CSD.)

Illustration 1: Main PNF Screen



NOTICE

ALL settings made in the PNF Tool are saved in the system save data. If a load from cold should need to be performed, the PNF settings will be restored during the system restore.

1.2 Firewall State



NOTICE

The Firewall should be ON at all times, this is the default setting. If issues are found because of the Firewall, contact the OLC connectivity team to resolve issues. It is not acceptable to leave the firewall OFF.

The Firewall can be turned ON and OFF by clicking the toggle button in [Illustration 2](#).

Illustration 2: Firewall State



1.3 Firewall Settings

The *Firewall Settings* toggle buttons are common across all tabs in the PNF tool. See [Illustration 3](#). These buttons can be clicked at any time and in any tab. The function of that button will be applied by the tool at the current state of **ALL** tabs.

Illustration 3: Firewall Settings Buttons



If there have not been any changes made the [Apply] and [Revert] buttons will not be active. The buttons will then appear as shown in [Illustration 4](#).

Illustration 4: Firewall Settings Buttons Inactive



Apply

The PNF Tool saves the current filter settings (from all Tabs). If the firewall is currently "On", the application will restart the firewall with these new settings. The [Apply] button appears flat & grayed out when PNF detects that the current GUI settings are the same as those last applied.

Restore

The PNF Tool changes the GUI filter settings to those last saved with the [Backup] button. If the firewall is currently "On", the application will restart the firewall with these new settings.

Factory Defaults

The [Factory Defaults] button will reset the PNF configuration defaults. If the firewall is currently "On", the application will restart the firewall with these new settings.

Backup

The PNF Tool saves a copy of the filtering rules that were last applied via the [Apply] button. The firewall is not restarted.

Revert

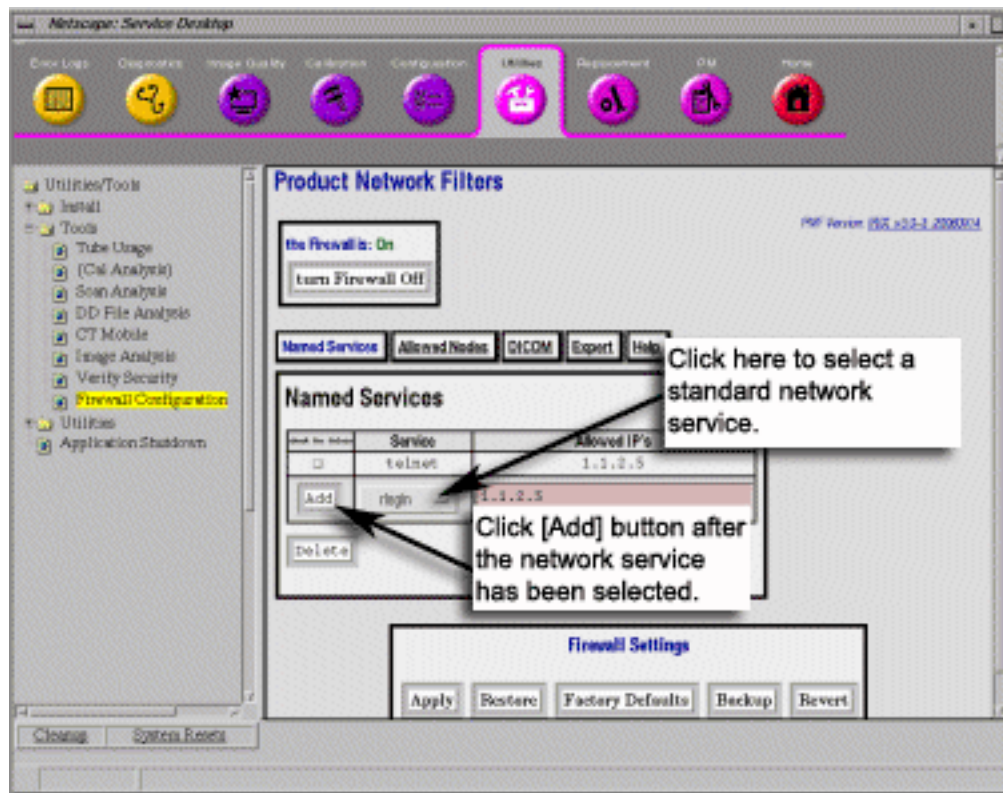
The PNF Tool changes the GUI filter settings to those last made when the [Apply] button was clicked. The firewall is not restarted. The [Revert] button appears flat & grayed out when PNF detects that the current GUI settings are the same as those last applied.

1.4 Tabs

1.4.1 Named Services Tab

The *Named Services* Tab, is the screen where the site's network administrator selects a standard networking service to open to this system from the site's network. See [Illustration 5](#).

Illustration 5: Named Services Tab



Adding a Network Service

- To add a network service, click the drop-down menu labeled *Services*. Standard Networking Services offered are:
 - telnet
 - ssh
 - ftp
 - rexec
 - rlogin
 - rsh
 - http
 - https
 - tftp
 - ldap
 - ssl/tls

- ntp
 - portmap
 - ping
2. Select a network service from the drop-down menu.
 3. An IP address can be entered in the *Allowed IP's* field or the field can be left blank. IP addresses allow the user to create rules where network communication is only accepted from a predetermined host or set of hosts. Below is a list of acceptable ways to populate the IP fields. The default value for IP fields is all IP addresses.
 - w.x.y.z = single IP address (ex: 1.2.34.5)
 - w.x.y.z/a.b.c.d = subnet with netmask (ex: 1.2.0.0/255.255.0.0)
 - w.x.y.z/a = subnet with byte netmask, this is the same as the previous subnet but specifying the netmask as a single byte (ex: 1.2.0.0/16)
 - Unlike PNF-Linux, the following syntax is not supported:
w.x.y.a - w.x.y.b = IP range, all hosts between two end points (ex: 1.2.34.5 - 1.2.34.10)
 4. Click [Add]. This service is now added to the list. If the *Allowed IP's* field was left blank an * will appear.
 5. Once all rules have been made, click the [Apply] to make changes to the PNF configuration.

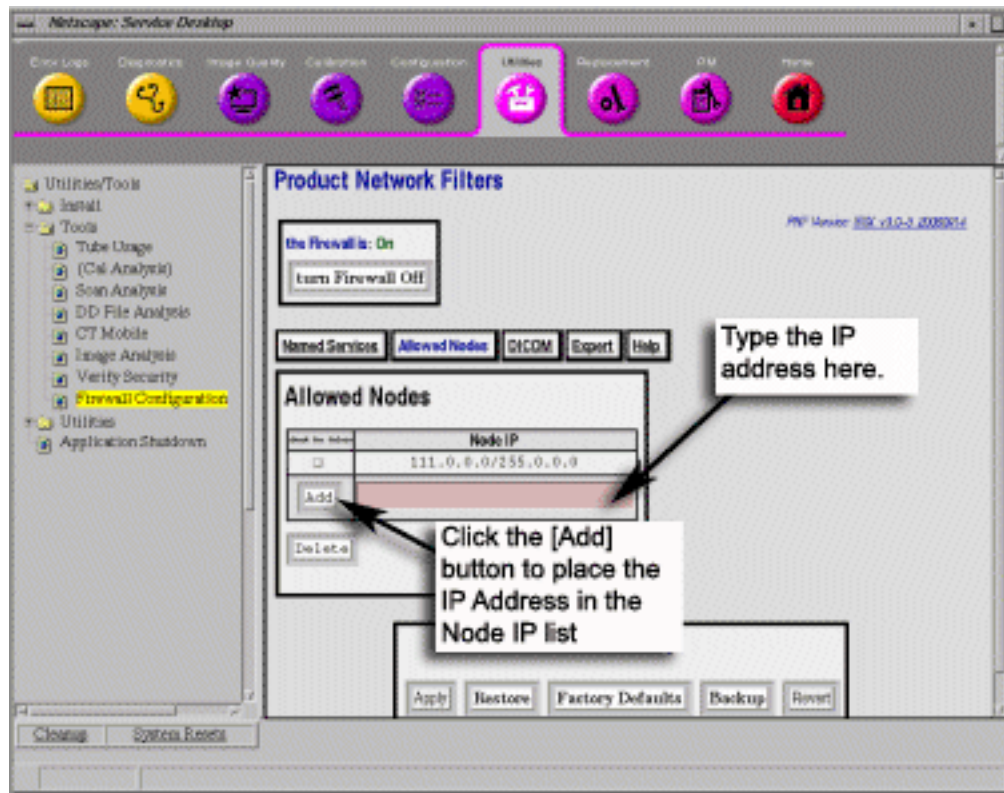
Removing a Network Service

1. Check the box next to the *Service* on the list.
2. Click the [Delete] button.
3. Once all rules have been made, click the [Apply] to make changes to the PNF configuration.

1.4.2 Allowed Nodes Tab

The *Allowed Nodes* tab opens all network services to a specific IP address. See [Illustration 6](#).

Illustration 6: Allowed Nodes



Adding an Allowed Node

1. Enter the IP address in the *Node IP* field.
IP addresses allow the user to create rules where network communication is only accepted from a predetermined host or set of hosts. Below is a list of acceptable ways to populate the IP fields. The default value for IP fields is all IP addresses.
 - w.x.y.z = single IP address (ex: 1.2.34.5)
 - w.x.y.z/a.b.c.d = subnet with netmask (ex: 1.2.0.0/255.255.0.0)
 - w.x.y.z/a = subnet with byte netmask, this is the same as the previous subnet but specifying the netmask as a single byte (ex: 1.2.0.0/16)
2. Once the IP address has been entered, click [Add] so the IP Address will be added to the list.
3. Once all Nodes have been added, click the [Apply] button to make changes to the PNF configuration.

Removing an Allowed Node

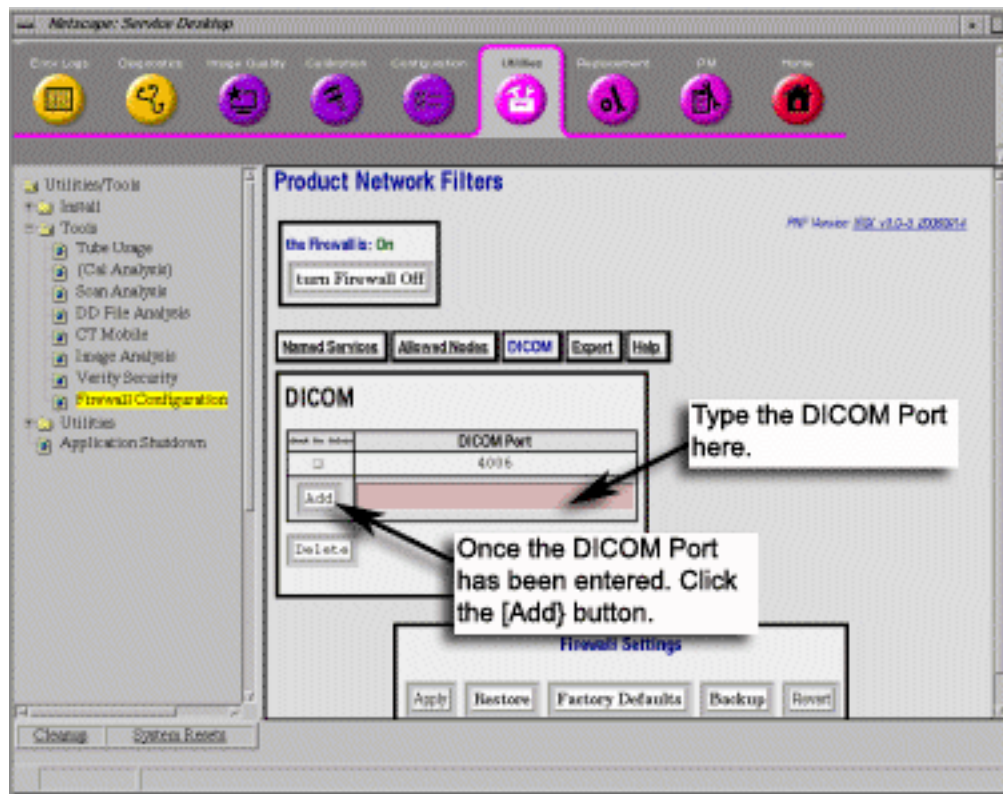
1. Check the box next to the *Node IP* in the list.
2. Click the [Delete] button to remove the *Node IP* from the list.

3. Once all changes have been made, click the [Apply] button to make changes to the PNF configuration.

1.4.3 DICOM Tab

This screen may be used to verify that the correct DICOM port was populated during installation. This field should be auto-populated during the initial set-up of the DICOM ports. Also additional DICOM ports can be added. See [Illustration 7](#).

Illustration 7: DICOM Tab



Adding a DICOM Port

1. Enter the DICOM Port in the field next to the [Add] button.
2. Click the [Add] button so the new DICOM Port is in the list.
3. Once all DICOM Ports have been added, click the [Apply] button to make changes to the PNF configuration.

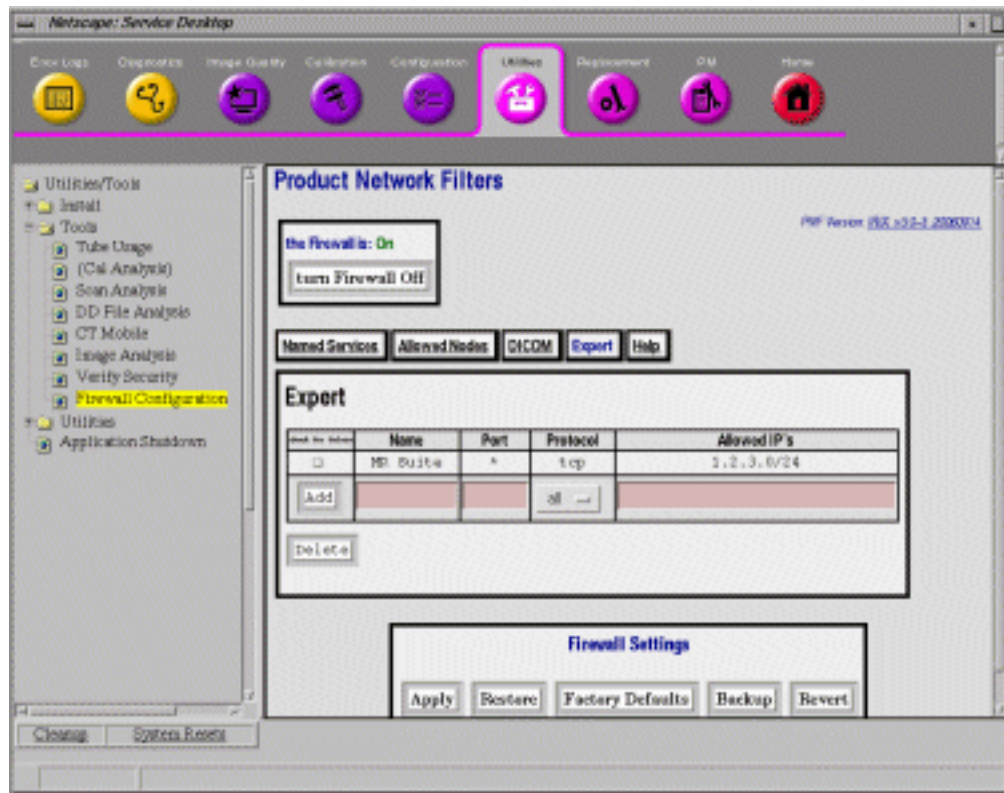
Removing a DICOM Port

1. Check the box next to the *DICOM Port* in the list.
2. Click the [Delete] button to remove the *DICOM Port* from the list.
3. Once all changes have been made, click the [Apply] to make changes to the PNF configuration.

1.4.4 Expert Tab

The *Expert* Tab allows a more flexible method for the network administrator to create rules. Any input to this tab will require networking expertise, please work with the site's network administrator when working in this tab.

Illustration 8: Expert Tab



Creating a Rule

1. To create a rule for the firewall, enter the application name in the *Name* field.
2. Enter the Port info in the *Port* field.
3. Click on the *Protocol* drop-down menu for the rule to be applied.
4. If needed, enter the IP address that this rule will be applied.
IP addresses allow the user to create rules where network communication is only accepted from a predetermined host or set of hosts. Below is a list of acceptable ways to populate the IP fields. The default value for IP fields is all IP addresses.
 - w.x.y.z = single IP address (ex: 1.2.34.5)
 - w.x.y.z/a.b.c.d = subnet with netmask (ex: 1.2.0.0/255.255.0.0)
 - w.x.y.z/a = subnet with byte netmask, this is the same as the previous subnet but specifying the netmask as a single byte (ex: 1.2.0.0/16)
5. Once all fields have been populated, click the [Add] button. The rules created will now be added to the list.

6. Once all rules have been made, click the [Apply] to make changes to the PNF configuration.

Removing a Rule

1. Check the box next to the *Name* in the list.
2. Click the [Delete] button to remove the rule from the list.
3. Once all changes have been made, click the [Apply] to make changes to the PNF configuration.

1.5 Troubleshooting

If the Firewall is suspected of causing an issue. A method to check if the Firewall is the source, turn the Firewall OFF and see if the issue remains. Make sure that the Firewall is ON before exiting. See [Illustration 9](#).

Illustration 9: Firewall State Button



The Firewall can be turned ON and OFF by clicking the [turn Firewall Off] or [turn Firewall On] buttons.



NOTICE

The Firewall should be ON at all times, this is the default setting. If issues are found because of the Firewall, contact the OLC connectivity team to resolve issues. It is not acceptable to leave the firewall OFF.

1.6 Finalization

Once all the necessary changes/settings have been made, perform the following steps:

1. At the *Firewall Setting* buttons, click [Apply]. See [Illustration 10](#) . This will apply the changes to the PNF configuration.
2. In order to store this configuration in the system data, click the [Backup] button. See [Illustration 10](#) . If a load from cold should need to be performed, the PNF configuration settings will be restored from the SaveInfo media.

Illustration 10: Firewall Setting Buttons



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